

ABINASH RAJASAGARAM

Selangor, Malaysia
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Portfolio:

CAREER OBJECTIVE

Bachelor of Software Engineering student at Universiti Putra Malaysia with a current CGPA of 3.80, seeking a Software Engineering internship from March to September 2027. Experienced in full-stack web development, backend systems, AI-powered applications, IoT integration, database management, REST APIs, and automation. Eager to apply technical and problem-solving skills to develop reliable, scalable, and innovative software solutions.

TECHNICAL SUMMARY

Hands-on experience developing full-stack web applications, AI-powered systems, IoT solutions, and database-driven applications. Familiar with frontend and backend development, REST API integration, authentication, database design, AI/ML integration, and testing. Project experience includes a food rescue platform, AI-powered hazard detection application, attendance management system, and AI event recommendation system. Additional exposure to IoT-based mushroom yield forecasting integrating environmental monitoring, computer vision, and machine learning. Strong interest in backend development, artificial intelligence, IoT, and scalable solutions.

SKILLS

Programming: JavaScript, Java, Python, SQL, HTML, CSS, Dart

Web Development: React, Vite, Node.js, Express.js, FastAPI, REST APIs

Database & Data Tools: PostgreSQL, Prisma ORM, SQL, Pandas, NumPy

Development Tools: Git, GitHub, VS Code, npm

IoT & Data Analytics: ESP32, IoT Sensors, Sensor Data Monitoring, Data Preprocessing, Feature Extraction

AI & Machine Learning:YOLOv8, Scikit-learn, Random Forest, Computer Vision, Image Segmentation, Embeddings, Cosine Similarity

Soft Skills: Communication, Problem Solving, Team Collaboration, Leadership, Adaptability

Languages: English, Malay, Tamil

Security & Authentication: JWT, Google OAuth, OTP Authentication, Role-Based Access Control

EDUCATION

Universiti Putra Malaysia | Bachelor of Software Engineering (Hons) — **October 2023 - Present**
Current CGPA:

Selangor Matriculation College | Physical Science — **July 2022 - May 2023**
CGPA: 4.00

SMK Pusat Bandar Puchong (1) | Pure Science — **Jan 2016 - March 2021**
Result:9A 1B

PROJECT EXPERIENCE

AI-Assisted Oyster Mushroom Yield Forecasting System (Final Year Project) — **2026**

- Developing an AI-assisted yield forecasting system integrating IoT environmental monitoring, computer vision, and machine learning.
- Implementing YOLO-based segmentation for mushroom growth analysis and Random Forest regression to predict harvest yield.
- Technologies: Python, ESP32, IoT Sensors, YOLO, Scikit-learn, Machine Learning, Computer Vision, Web Dashboard.

AI-Powered Hazard Detection Mobile Application (https://github.com/axxnash/yolomodel_hazard.git) — **2026**

- Developed an AI-powered mobile application to detect workplace and facility hazards from uploaded or captured images.
- Integrated multiple YOLOv8 models for hazard detection, confidence scoring, severity classification, and AI-generated safety recommendations.
- Technologies: Flutter, Dart, Spring Boot, Java, Python, FastAPI, YOLOv8, Gemini API.

FEEDBACK | Food Rescue and Distribution Platform (https://github.com/axxnash/FeedBack--.git) — **2026**

- Developed a multi-role platform connecting vendors, food recipients, NGOs, and delivery riders to reduce food waste.
- Implemented discounted food listings, food donation requests, vendor offers, NGO verification, delivery management, and QR/OTP pickup verification.
- Technologies: React, JavaScript, Node.js, Express.js, PostgreSQL, Prisma ORM, JWT.

AI Event Recommender System for Students (https://github.com/axxnash/ai-project.git) — **2026**

- Developed an AI-powered system that recommends relevant campus activities based on student preferences.
- Implemented semantic similarity and embeddings-based ranking for personalised event recommendations.
- Technologies: React, Vite, Python, FastAPI, PostgreSQL, Scikit-learn, Pandas, NumPy.

LEADERSHIP & ENGAGEMENT

Head of Technical Department | University & Inter-University Events — **2024 - Present**

Led technical operations, audiovisual setup, troubleshooting, and equipment management for university and inter-university events.

Exco, Public Relations & Volunteering | PHUPM — **2025 - 2026**

Led 30 student volunteers, coordinating community outreach, volunteer assignments, and charitable activities.

Programme Director | Aram Seivom — **2025**

Directed a charity programme, coordinating donations, fundraising, educational activities, and community engagement.

Multimedia Committee | Software Engineering Club, Faculty — **2024**

Managed social media, event photography, promotional content, and digital publicity for faculty activities.

REFERENCES

Ts. Dr. Ng Keng Yap
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